

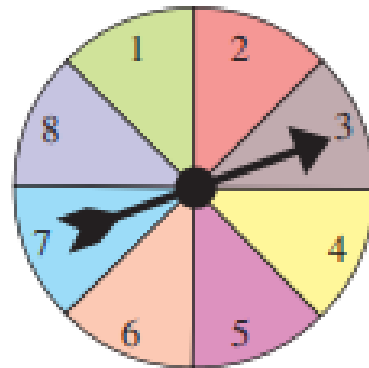
INTRODUCTION TO PROBABILITY THEORY

Chapter 1

INTRODUCTION TO PROBABILITY

TERMINOLOGY

- The circular spinner at right is divided into 8 equal sectors. When the spinner is spun, the possible outcomes are 1, 2, 3, 4, 5, 6, 7, 8.
- These outcomes may be listed as the elements of a set.
- The set of all possible outcomes of an experiment is called the **sample space** (or the universal set) and is denoted by $\mathfrak{E}$, and each possible outcome is called a **sample point**.



CONT.

- Therefore, spinning the spinner gives
$$\varepsilon = \{1, 2, 3, 4, 5, 6, 7, 8\}$$
- A subset of the sample space is known as an **event**.
- For the example above, if event A is defined as ‘an odd number when the spinner is spun’, then $A = \{1, 3, 5, 7\}$.
- If event B is defined as ‘a number less than 5 when the spinner is spun’, then $B = \{1, 2, 3, 4\}$.

CONT.

- The **union** (symbol $\cup$) of the two events A and B above implies a combined event, that is, either event A or event B or both occurring.

Therefore the set $A \cup B = \{1, 2, 3, 4, 5, 7\}$.

Note: Common elements are written only once.

- The **intersection** (symbol $\cap$) of the two events A and B above is represented by the common sample points of the two events.

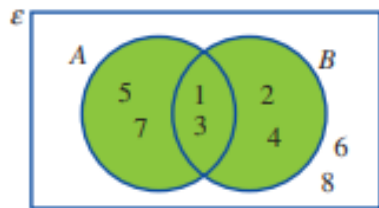
Therefore the set $A \cap B = \{1, 3\}$.

VENN DIAGRAMS

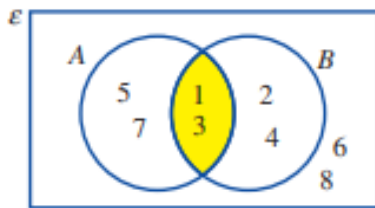
- Venn diagrams involve drawing a rectangle that represents the sample space and a series of circles that represent subsets of the sample space.
- They provide a visual representation of the information at hand and clearly display the relationships between sets.

CONT.

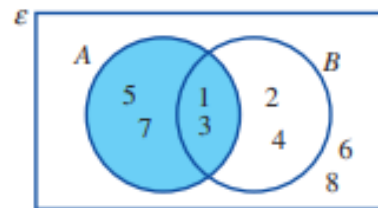
- The Venn diagrams below illustrate an alternative way of presenting information regarding the circular spinner shown on the previous slides.



■ $A \cup B$



■ $A \cap B$

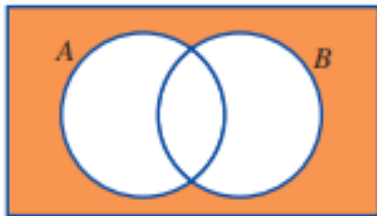


■ A

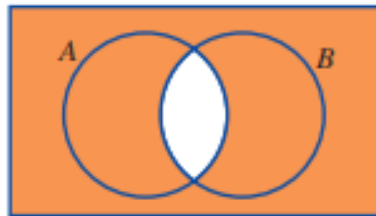
- Note: Sample points not belonging to either set are placed outside the circles but remain inside the rectangle.

CONT.

- Venn diagrams can also assist in determining whether or not two sets are equal, that is, whether they contain the same elements.
- As the examples below show, the equality of two sets may not be obvious from the set notation but is often easier to see in a diagram.



 $A' \cap B' = (A \cup B)'$



 $A' \cup B' = (A \cap B)'$

PROBABILITY

- Probability deals with the likelihood or chance of some event occurring. The probability of a specific event, say A , occurring is defined by the rule:

$$\Pr(A) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

- Its probability lies within the restricted interval $0 \leq \Pr(A) \leq 1$.
- A probability of zero implies that the event cannot occur, while a probability of 1 implies that the event will most certainly occur.

CONT.

- The individual probabilities of a particular experiment will sum to a value of 1 and can be denoted as follows.

$$\sum p(x) = 1$$

- If $\Pr(A)$ is defined as the probability of an event occurring, then its complement, $\Pr(A')$, is defined as the probability of an event not occurring.
- Therefore, it can be stated that $\Pr(A) + \Pr(A') = 1$ which can be transposed to $\Pr(A') = 1 - \Pr(A)$.

EXAMPLE 1

Two fair dice are rolled simultaneously and the sum of the two numbers appearing uppermost is recorded as shown below.

(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

- Find the probability that the sum will be:
 - a 6
 - b 10
 - c a number less than 5
 - d at least 9
 - e an odd number.

EXAMPLE 2

A bag contains 15 marbles comprising 5 black, 3 red, 4 blue, 2 white and 1 green. One marble is drawn randomly from the bag.

a Determine the probability of each of the coloured marbles being drawn:

- i black

- ii red

- iii blue

- iv white

- v green.

b Show that the probabilities sum to 1.

c What is the probability that the marble drawn is:

- i not black?

- ii either black or white?

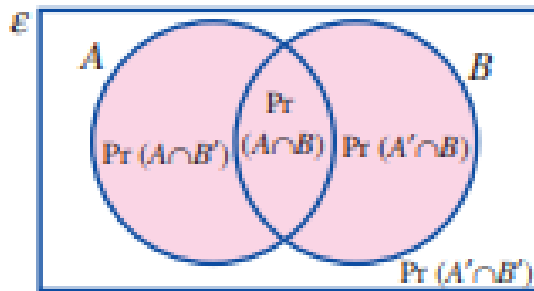
- iii neither blue nor green?

THE ADDITION RULE OF PROBABILITY

The addition rule of probability states that

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B).$$

Venn diagrams may also be used to display the probabilities rather than just the outcomes, as shown in the diagram below.



EXAMPLE 3

- a If $\Pr(A) = 0.4$, $\Pr(B) = 0.7$ and $\Pr(A \cap B) = 0.2$,
find $\Pr(A \cup B)$.
- b If $\Pr(A) = 0.6$, $\Pr(B) = 0.8$ and $\Pr(A \cup B) = 0.9$,
find $\Pr(A \cap B)$.

MUTUALLY EXCLUSIVE EVENTS

- If two or more events cannot occur simultaneously, they are said to be mutually exclusive or disjoint; that is, they have nothing in common. In set notation this may be expressed as

$$\Pr(A \cap B) = \{\} \text{ or } \Pr(A \cap B) = 0.$$

- Therefore, for mutually exclusive events the addition rule becomes:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B).$$

INDEPENDENT EVENTS

- Two events A and B are independent if one event does not influence the other event from occurring. The mathematical definition of independence is given by:

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B).$$

EXAMPLE 4

Two fair dice are rolled with S representing the event of obtaining a number less than 4 on the first die and T the event of obtaining a number greater than 4 on the second die. Find:

- a $\Pr(S)$
- b $\Pr(T)$
- c if events S and T are mutually exclusive
- d if events S and T are independent.

EXAMPLE 5

Two fair dice are rolled with U representing the event of obtaining a 5 on the first die and V the event of the sum of numbers on the two dice exceeding 10. Find:

- a $\Pr(U)$
- b $\Pr(V)$
- c if events U and V are independent.

KARNAUGH MAPS AND PROBABILITY TABLES

Karnaugh maps and probability tables summarise all combinations of two events (for example A and B) and their complements (for example A' and B').

	B	B'	
A	$A \cap B$	$A \cap B'$	
A'	$A' \cap B$	$A' \cap B'$	

Karnaugh map

	B	B'	
A	$\Pr(A \cap B)$	$\Pr(A \cap B')$	$\Pr(A)$
A'	$\Pr(A' \cap B)$	$\Pr(A' \cap B')$	$\Pr(A')$
	$\Pr(B)$	$\Pr(B')$	1

Probability table

EXAMPLE 6

		Column 1	Column 2	Column 3
		<i>B</i>	<i>B'</i>	
Row 1	<i>A</i>	0.12		
Row 2	<i>A'</i>			0.5
Row 3			0.63	1

For the probability table shown at right, event *A* is ‘not more than 17 years of age’ and event *B* is ‘has a learner permit’.

a Complete the probability table at right.

b What do the following probabilities represent?

i $\Pr(A \cap B)$

ii $\Pr(A' \cap B')$

c What is the probability that:

i a person over the age of 17 does not have a learner permit?

ii a person has a learner permit and is older than 17?

iii a person over the age of 17 has a learner permit or a person at or under the age of 17 does not have a learner permit?

CONDITIONAL PROBABILITY

- Conditional probability deals with an event which has previously occurred and has an effect on an event we are interested in. Due to the initial condition (or restriction) imposed, the number of possible events is reduced.
- Conditional probability is defined by the rule
$$\Pr(A|B) = \Pr(A \cap B) / \Pr(B), \text{ where } \Pr(B) \neq 0,$$
and can be transposed to
$$\Pr(A \cap B) = \Pr(A|B) \times \Pr(B).$$
- The latter is called the multiplication rule.
- $\Pr(A|B)$ is read as ‘the probability of A given B’.

EXAMPLE 7

- If $\Pr(A) = \frac{1}{5}$, $\Pr(B) = \frac{1}{10}$ and $\Pr(A \cap B) = \frac{1}{20}$,
find:
 - a $\Pr(A \cup B)$
 - b $\Pr(A | B)$
 - c $\Pr(B | A)$
 - d if events A and B are mutually exclusive
 - e if events A and B are independent.

TREE DIAGRAMS

- Tree diagrams are a useful tool in solving probability tasks as they display each of the possible outcomes along with their respective probabilities.

EXAMPLE 8

- Nadia knows that if her car starts, she has an 80% chance of getting to work on time. However, if her car doesn't start, her chance of arriving on time is 50%. If Nadia's car starts only 70% of the time, what is the probability that:
 - a her car starts and she gets to work on time?
 - b she arrives at work late?
 - c she arrives at work on time?
 - d her car starts, given that she arrives at work on time?

EXAMPLE 9

- A fair coin is tossed three times. Find the probability of obtaining two heads given the first toss resulted in a tail.

COMBINATIONS

- In mathematics, a combination deals with the number of ways items may be selected from a set of elements where the order is not important.
- For example, in how many ways can 3 numbers be selected from the set $\{1, 2, 3, 4\}$, taking into account that order is not important? The following selections can be made: 1, 2, 3 1, 2, 4 2, 3, 4 3, 4, 1
Hence 4 selections could be made.
- If order was important, there would be a greater number of possibilities since each of the above selections could be arranged in 6 ways. For example, the selection (1, 2, 3) could be arranged as:
1, 2, 3 2, 1, 3 2, 3, 1 3, 1, 2
3, 2, 1 1, 3, 2.
- A combination is also referred to as a selection or choice, and is defined by the rule nC_r .

nC_r = the number of selections of n different objects taken r at a time

$$= \frac{n!}{(n-r)!r!}$$

nC_r may also be expressed as $\binom{n}{r}$ and is read as ' n over (above) r '.

EXAMPLE 10

A drawer contains 7 T-shirts of which 3 are white and the rest are black. If 2 T-shirts are randomly selected from the drawer simultaneously, find the probability that they are:

- a both black
- b both white
- c different colours
- d the same colour.